

LSTM

26 August 2024 08:09

Problem with RNN
 19 December 2022 16:33

RNNs → sequential data → textual, time series

Suffer 2 major problems → **LSTMs** (Way)

→ Problem of long term dependency } → Unstable

→ Unstable gradients

timesteps → RNN

predict: Marathli is spoken in Marathli

short

→ Marathli is a beautiful place. I went there last year, but I could not enjoy it properly. I don't understand Marathli

Vanishing gradient prob → long term → RNN

Problem #1 → Problem of long term dependency

3 time steps

inp: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ out: $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

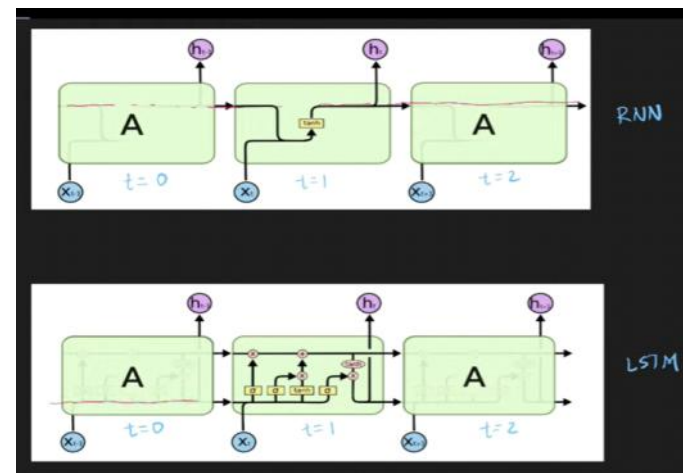
hidden: x_1, x_2, x_3

weights: w_1, w_2, w_3

output: $y = 1$

loss: $L = \min$

gradient des



3 time steps

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gradient des

long term dep

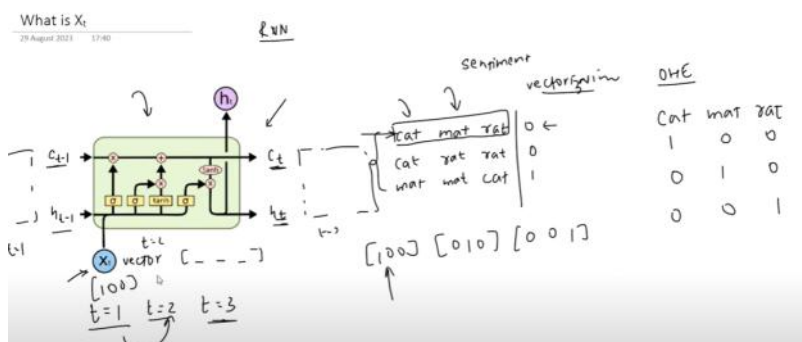
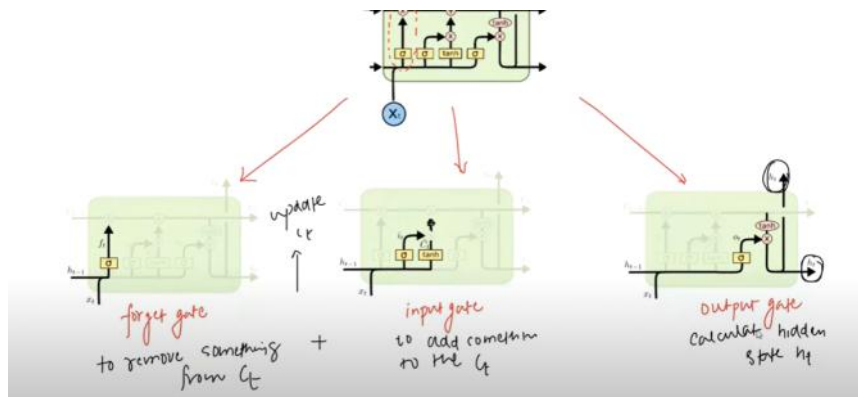
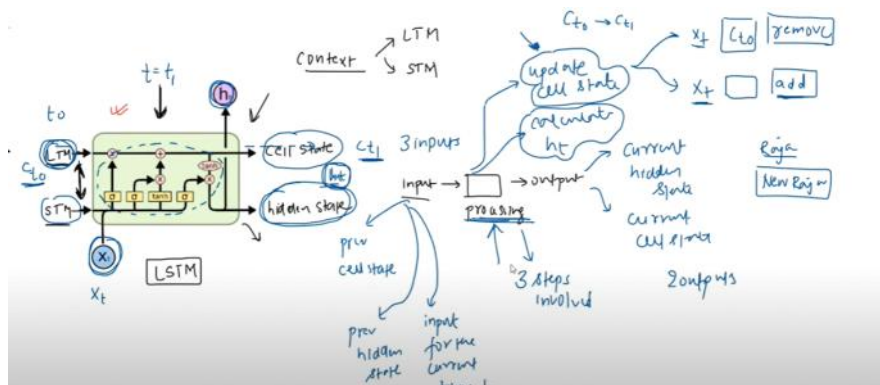
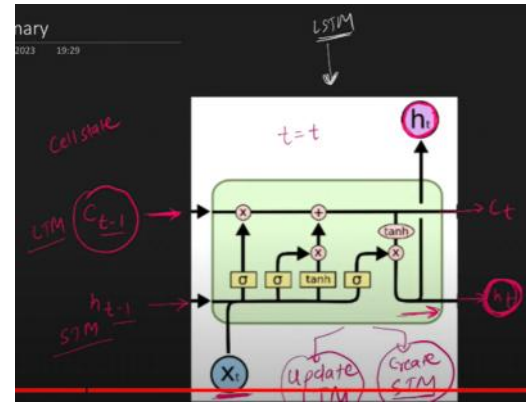
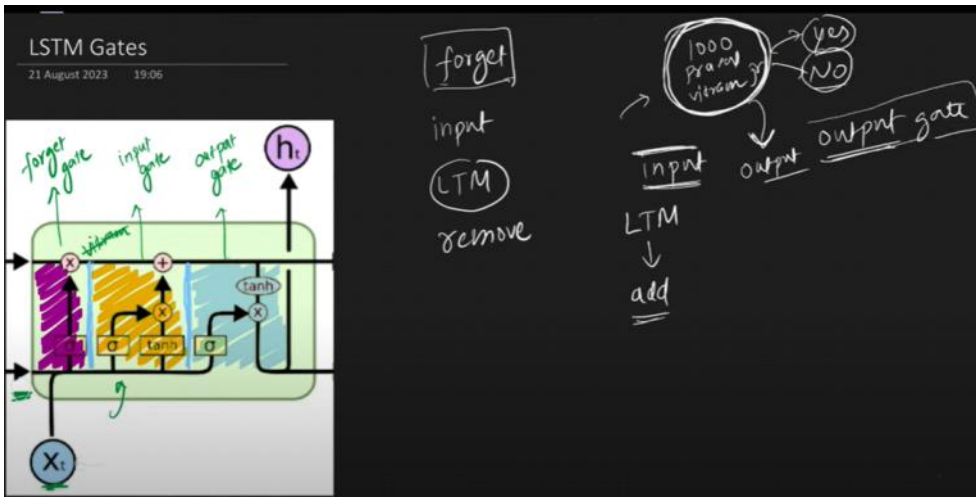
Soln

- 1) Diff activation → relu / leaky relu
- 2) Better weight init
- 3) skip nnns
- 4) **LSTM**

Problem #2 → Unstable Training (Exploding gradients)

- 1) Gradient clipping
- 2) Controlled learning rate
- 3) **LSTM**

LSTM



Ct, Ht are vectors and the dimensions for both of them are same and their dimension will be same as number of nodes in the neural network. Xt is one word and a vector (the dimension can vary from Ct, Ht)

What are f_t , i_t , o_t and $\bar{C}_t$

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forget gate

input gate

output gate

candidate cell state

vectors

$[x \ y \ z]$

$[\quad \quad]$

f_t i_t $\bar{C}_t$ o_t

f_t , i_t , $\bar{C}_t$, o_t are of the same dimension vectors of C_t , H_t

Pointwise Operations

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$\otimes$

$+$

$\tanh$

$C_{t-1} = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} \rightarrow$

$f_t = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

$C_{t-1} \otimes f_t \rightarrow \text{vector}$

$\rightarrow \begin{bmatrix} 4 & 10 & 18 \end{bmatrix}$

$\begin{bmatrix} 5 & 7 & 9 \end{bmatrix}$

X = Element wise /point wise multiplication
Pointwise tanh - passing each vectors through tanh and getting the values

Neural Network Layers

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ANN

hidden

num units nodes

hyperparameter

forget

tanh

Number of nodes inside the neural network is a hyperparameter and once decided the number of nodes or perceptron will be same for all the 4 neural networks.

The Forget Gate

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vectors dim

dim

input

output $\rightarrow$ cell state

remove

calculate f_t

$C_{t-1} \otimes f_t$

$f_t \rightarrow [f_1 \ f_2 \ f_3] \rightarrow f_t (3)$

$b_1 \ b_2 \ b_3 \rightarrow (1 \times 1) \ (3 \times 7) \ (7 \times 1)$

$\rightarrow (3 \times 1) + (3 \times 1) = (3 \times 1)$

$W_{3 \times 7}$ (21 weights)

bias

21 wts

4 biases

$h_{t-1} (3)$

$x_t (4)$

7×1

C_{t-1} and H_{t-1} dimension will be same as number of nodes in the neural network. E.g here is 3 and X_t dimension is 4 dimension vector

W_f = total weights . H_{t-1} , X_t = Concatenation of vectors H_{t-1} and X_t

f_t

$f_t = W_f [h_{t-1} \ x_t] + b_f$

(3×1)

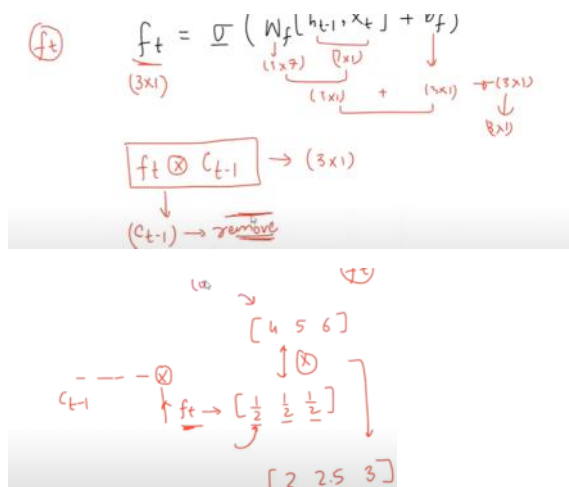
(3×7)

(7×1)

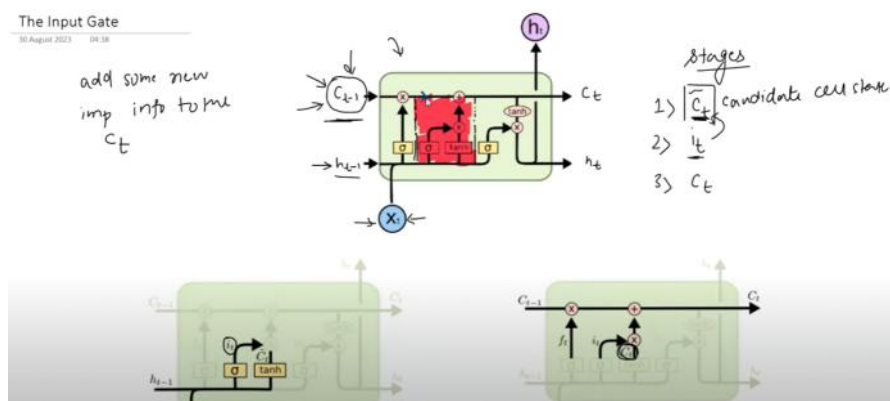
$(3 \times 1) + (3 \times 1) \rightarrow (3 \times 1)$

$f_t \otimes C_{t-1} \rightarrow (3 \times 1)$

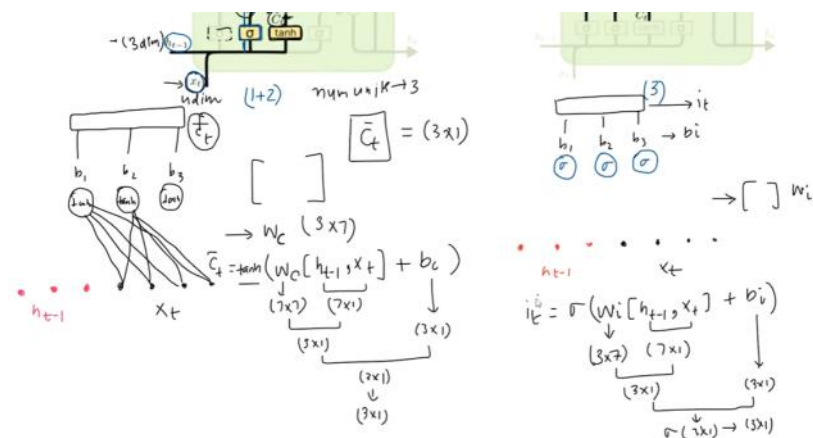
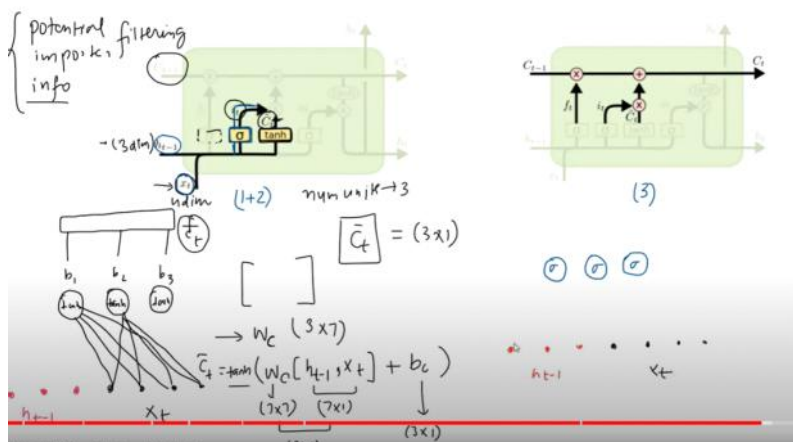
$(C_{t-1}) \rightarrow \text{remove}$



We took C_{t-1} and we removed some of the information



It decides from the potential candidate cell state which needs to be added
 C_t = new Cell state ($f_t * C_{t-1}$)



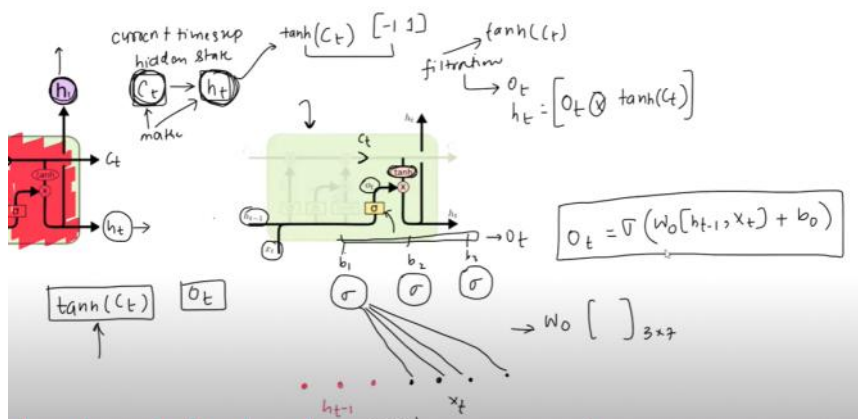
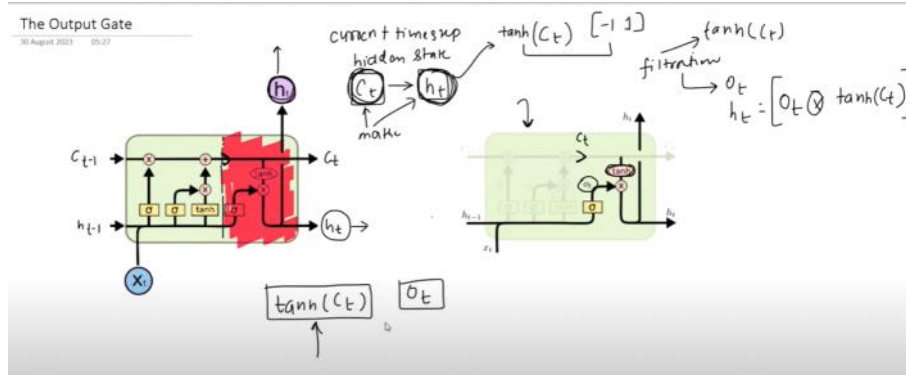
It is filter which will decide what should be taken as a cell state from the candidate cell state

pointwise
 $i_t \otimes \bar{c}_t \rightarrow \bar{c}_t^*$ (filtered candidate cell state)
 $(3 \times 1) \quad (3 \times 1)$

$\begin{bmatrix} 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$
 $\downarrow$ 50% or more
 $\begin{bmatrix} 2 & 2.5 & 3 \end{bmatrix}$

$$C_t = f_t \otimes C_{t-1} \oplus i_t \otimes \bar{C}_t$$

c_t



Step 1: $\tanh(C_t)$
 Step 2: Filtration (O_t)
 Step 3: $h_t = O_t \otimes \tanh(C_t)$

$x_t \rightarrow (3 \times 1)$ $\rightarrow (3 \times 1)$

$\rightarrow O_t$ $O_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$

$\rightarrow W_o$ 3×7 (7×1) (3×1) (3×1)

$\rightarrow W_o$ 3×7 (3×1) (3×1)

$h_t = O_t \otimes \tanh(C_t) \rightarrow$

$(3 \times 1) \quad (3 \times 1)$

Ct: Cell state:
 Maintains long-term memory, and stores information over time.
 Represents the memory of the LSTM, retaining information across long sequences.
 Indirectly affects the hidden state through gate operations.
 Preserves information across multiple time steps.

Ht (Hidden state):
 Captures short-term dependencies, and reflects the current state of the network.
 Carries information passed to the next time step, influencing network predictions.
 Directly contributes to the hidden state, which is the output of the LSTM cell.
 Reflects the current state of the network's learning process at each time step.

From <https://www.geeksforgeeks.org/whats-the-difference-between-the-cell-and-hidden-state-in-lstm/>